Nonblood Specimen Collections

As a health care professional, you may be tasked with collecting various specimens or instructing patients on how to collect them at home. Common specimens include urine, stool, sputum, and semen. Nonblood specimens have different handling requirements. Phlebotomists need to have a thorough understanding of these protocols to maintain the quality of the sample and guarantee precise outcomes.

Urine Collection

Routine urinalysis is one of the most frequently conducted laboratory tests, analyzing a patient's urine for characteristics such as appearance, color, smell, pH level, specific gravity, and the presence or absence of various substances like protein, glucose, and hemoglobin. Urinalysis may be performed on a random sample or collected at a specific time, such as the first urine of the day.

Another prevalent test is a urine culture and sensitivity, which involves a lab technician preparing the urine sample to cultivate and identify micro-organisms. This is often followed by a sensitivity test to determine the effectiveness of antibiotics against the identified pathogen. For these tests, a clean-catch (midstream) urine sample with a sterile container is necessary, or the sample may be obtained via a catheter. It's important to provide patients with both verbal and written instructions to ensure they collect the urine sample correctly for culture and sensitivity testing.

Certain assessments, such as those evaluating kidney function, require a timed urine sample. This process entails collecting all urine output over a set period, which could be 2, 4, or, most commonly, 24 hours.

Random Urine Specimen

These specimens can be collected at any time. Provide the patient with a clean, disposable container and a label with their information. Instruct them to urinate directly into the container, collecting 30 to 60 mL of urine. If collected at home, the specimen should be refrigerated and brought to the lab promptly, with the collection date and time noted on the label.

First Morning Urine Specimen

These specimens are collected by the patient first thing in the morning. This specimen is typically more concentrated, with a higher specific gravity, and is used for specific tests. A concentrated urine specimen can contain substances and elements, such as cells and cast that would not typically be detected in a more dilute random specimen. In an outpatient setting, the patient should refrigerate the sample and deliver it to the lab as soon as possible.

Clean-Catch (Midstream) Urine Specimen

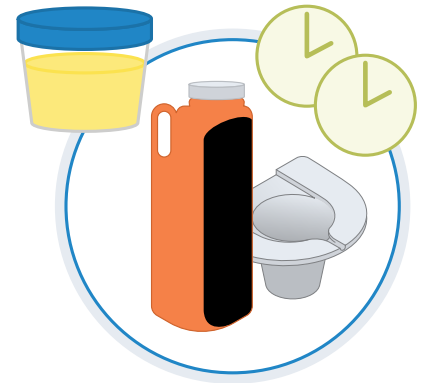
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Male Patients: Instruct patients to clean from the meatus outward in a circular motion using a sterile antiseptic wipe. If uncircumcised, the foreskin should be retracted and cleansed using the same technique. Dispose of the antiseptic wipe and repeat the process with a new sterile antiseptic wipe. After cleansing, the patient should begin to void a small amount of urine into the toilet, then collect urine in the container until it is one-fourth to one-half full (30 to 100 mL), keeping the penis from touching the container. Once the sufficient amount is collected, void the remaining urine into the toilet.

Female Patients: Instruct the patient to separate the labia around the urinary meatus. Using sterile antiseptic wipes, clean each side and down the middle, using a fresh wipe for each area and wiping from front to back. The patient will begin voiding a small amount into the toilet, then collect the urine in the container until it is one-fourth to one-half full (30 to 100 mL), only touching the outside of the container. Once the sufficient amount is collected, void the remaining urine into the toilet.

Timed Urine Specimen

When guiding a patient on how to collect a timed urine sample, supply them with a suitable container, which may contain a preservative if required by your facility. Prior to providing the container to the patient, attach a label stating the patient's identification information. These containers are typically sturdy, opaque to light, have a capacity of around 3,000 mL, and are equipped with a wide opening and a secure, leakproof screw-on lid. To start a 24-hour urine collection, the patient should note the exact time, empty their bladder into the toilet, and then begin collecting all subsequent urine in the provided container until the same time the next day. It's crucial to emphasize that if any urine is accidentally passed into the toilet and not the container during the 24-hour window, the collection is compromised and must be restarted. Throughout the collection period, the urine should be stored on ice or refrigerated. Make sure the patient understands that the container should contain only urine. The final urine void should be added to the container as close to the 24-hour mark as possible.



Urine Reagent Testing

In certain health care settings, phlebotomists may visually assess the specimen for color, clarity, and odor, along with carrying out point-of-care urine testing using reagent strips, which is considered a CLIA-waived test. This testing method involves immersing a chemically treated strip into a urine sample. The chemical analysis is done by observing the color changes on the strip at specific time intervals and recording the findings. This can be done with an automatic machine or manually. Proper procedure dictates wearing clean gloves, inserting the reagent strip into the urine, monitoring the test areas for color shifts, and comparing these to the reference chart on the reagent container.

Follow Manufacturer's Instructions

It's important to follow the manufacturer's instructions closely, as the color readings can be affected by excess urine on the strip or incorrect timing when interpreting the results.

Reagent strips typically measure several components of urine.

- **pH:** The pH level can range from about 5 to 7, depending on diet. pH is relevant for identifying bacterial growth, as bacteria thrive more in alkaline conditions. Unrefrigerated samples may become more alkaline over time, leading to false elevations in bacterial counts.
- **Specific gravity:** This measures urine concentration, with expected values ranging from 1.003 to 1.030, though they typically fall between 1.010 and 1.025. Values below 1.010 suggest dilute urine, while values above 1.010 indicate concentrated urine, which could signal dehydration.
- **White blood cells and nitrites:** These are typically not found in urine. The presence of these elements may indicate an infection.
- **Ketones:** These are byproducts of fat metabolism and may be present due to conditions like diabetes, fasting, or excessive vomiting.
- **Bilirubin:** Elevated levels can suggest liver issues or the breakdown of red blood cells.
- **Protein:** Protein is typically not found in urine. Protein in urine can be a marker for inflammation, infection, kidney disease, or toxicity.
- **Glucose:** Glucose is typically not found in urine. The presence of glucose may be an indicator of diabetes.

All other substances tested by the urine reagent strip are expected to be negative in a standard urine sample. While reagent strips are useful for initial screening, any unexpected results typically require further laboratory testing for confirmation. Additional information on urine testing can be found in *Chapter 5: Special Collections*.

results of a urinalysis



quantitative

glucose	negative
specific gravity	1.020
pH	6.1
protein	negative
ketones	negative
WBCs	negative
bilirubin	negative
blood	negative
nitrate	negative
urobilinogen	0.3

qualitative

clarity	clear
color	amber

Urine Drug Screening

Urine drug screening is performed to detect illegal use of recreational drugs, steroids to increase performance, and the use of prescription medications. A random sample can be collected to perform a urine drug screening. Specific chain of custody requirements are required for these tests. Depending on the reason for the testing, strict collection guidelines may be necessary. A chain of custody form includes the date, time of collection, place where the specimen was obtained, and signature from the individual whom the specimen was obtained from. This form is used to maintain the integrity of the sample by providing a documented trail that shows it has been handled properly and has not been tampered with.

Stool Collection

The ***fecal occult blood test (FOBT)*** is a widely used point-of-care test that detects hidden blood in the stool, aiding in the diagnosis of gastrointestinal issues like ulcers and colorectal cancer. Additionally, stool samples are analyzed to identify the presence of bacteria, viruses, fungi, and parasites.

When instructing a patient on how to collect a stool sample for lab testing, provide them with a sterile, wide-mouth plastic container with a secure lid. This container is designed for the patient to pass stool directly into it. Alternatively, patients can use a clean receptacle to collect the stool and then transfer it to the provided container using a tongue depressor or a built-in scoop. It's important to advise patients to prevent urine or toilet water from contaminating the stool sample, as this can inhibit the growth of micro-organisms being tested for. Only a small quantity of stool is needed for the test. After collecting the sample, the patient should tightly close the container, clean its exterior if needed, and wash their hands thoroughly. Stool samples for parasite testing should not be refrigerated and must be taken to the lab immediately. Preserved specimens can generally be kept at room temperature, but other types of stool samples, including multiple-day stool collections, may need to be refrigerated and brought to the lab later for analysis.

For FOBT, patients are given a kit with special cards, instructions, and a return envelope for sending the samples to the lab. The instructions typically require the collection of three separate areas of the stool sample, which are then smeared onto the cards using the provided applicators. Samples should not be taken from toilet water; collecting them in a container or directly from toilet tissue is recommended.

When discussing stool collection, it's essential to maintain a professional attitude and provide clear, straightforward guidance, while also encouraging patients to ask any questions they may have.

fecal occult blood test (FOBT). A lab test used to check stool samples for hidden (occult) blood.

Pretest Requirements

Patients are advised to avoid certain foods like red meat, citrus fruits, and raw vegetables, as well as medications like vitamin C and aspirin for 3 days prior to sample collection. This will help avoid potential interference with the test results from substances that can mimic the presence of blood and cause false results.

Sputum Collection

Sputum, which is expelled from the lungs when a person coughs, is often tested to identify specific pathogens causing respiratory infections. It can also be used to assess the effectiveness of a treatment regimen. Because sputum can carry infectious agents, it's important to handle it with care.

For the most accurate test results, patients should provide a sputum sample early in the morning before consuming any food or beverages. This timing ensures that the sputum is undiluted and contains a higher concentration of potential pathogens. Eating or drinking could dilute the sputum, making it less useful for diagnostic purposes. The patient is provided with a sterile container for the sputum collection.

The patient should be guided to take a deep breath and then cough forcefully to bring up sputum from the lungs, then expel the sputum directly into the container, taking care to avoid contaminating the sample with saliva. Typically, 1 to 2 tsp of sputum are needed for an adequate sample. The lid should be securely closed and the exterior wiped clean and labeled properly. To preserve specimen quality, it should remain at room temperature and be transported to the laboratory promptly for processing.

Sputum Collection

The patient should be instructed to avoid using mouthwash, eating, or drinking before collecting a sputum sample, as mouthwash can destroy some of the micro-organisms present in the sputum.

Semen Collection

Semen analysis is a laboratory procedure conducted for various reasons, including fertility assessments, postvasectomy evaluations, or forensic analysis in criminal cases. The specific purpose of the semen test may require the patient to follow certain precollection instructions, such as refraining from sexual activity or avoiding alcohol for a set period before providing the sample.

To ensure an accurate analysis, it is essential to use a collection container that is sterile and devoid of any contaminants like detergents or spermicides. The patient should ejaculate directly into the provided container, taking care to avoid any spillage or contamination. Once the sample is collected, the lid should be tightly secured to prevent leakage. The container should be labeled with the patient's name, the date, and the exact time of collection. To maintain the integrity of the sample, it must be kept warm and should be kept away from extreme temperatures, both hot and cold. The semen sample should be delivered to the laboratory ideally within 1 hour of collection for optimal results.

By adhering to these guidelines, the patient can help ensure that the semen sample is properly collected and preserved for accurate laboratory analysis.

Conclusion

The meticulous attention to pretest preparations and specimen collection is paramount in achieving accurate and reliable diagnostic results. By effectively communicating with patients, confirming their adherence to precollection instructions, and addressing any special considerations, phlebotomists can minimize the risk of compromised samples and ensure patient safety. Additionally, understanding instructions of nonblood specimen collections and providing patient education on procedures like venipuncture are essential in fostering a supportive and informed environment. With empathy, clear communication, and a thorough understanding of protocols, we can enhance the patient experience and contribute to the overall success of diagnostic testing in health care.

3.10 CHALLENGE

1. Match the type of urine test with its description.
   1. Random urine
   2. 24-hour urine collection
   3. Postprandial
   4. Clean-catch midstream
   5. Glucose tolerance

   A. A method of performing a urine collection that helps prevent bacterial contamination
   B. A urine specimen that may be collected at any time without the restrictions of fasting, timing, or aseptic condition
   C. Performed at the same time as blood tests every 1 to 2 hr; patient must be fasting
   D. The patient receives a container (with a lid) large enough to hold 3 to 4 liters of fluid. Instruct the patient to void and discard the first morning urine specimen, then initiate the collection process after discarding that morning specimen.
   E. Requires a patient to void and then eat a meal; collected 2 hr after a patient eats a prescribed amount of carbohydrates or glucose solution

2. Match the type of specimen collection with its corresponding protocol and rationale.
   1. Routine urine collection
   2. Clean-catch (midstream) urine
   3. Timed urine specimen

   A. The patient should urinate a small amount into the toilet, then collect midstream urine in a sterile container without touching the inside.
   B. The patient can collect the sample at any time in a clean, disposable container and, if the sample is collected at home, should refrigerate it until it can be brought to the lab.
   C. The patient should collect all urine over a specified period, such as 24 hours, and keep the sample refrigerated or on ice.

3.10 CHALLENGE KEY

1. 1 = B. 2 = D. 3 = E. 4 = A. 5 = C.
2. 1 = B. Routine urine collection: The patient can collect the sample at any time in a clean, disposable container and, if the sample is collected at home, should refrigerate it until it can be brought to the lab.

   2 = A. Clean-catch urine: The patient should urinate a small amount into the toilet, then collect midstream urine in a sterile container without touching the inside.

   3 = C. Timed urine specimen: The patient should collect all urine over a specified period, such as 24 hours, and keep the sample refrigerated or on ice